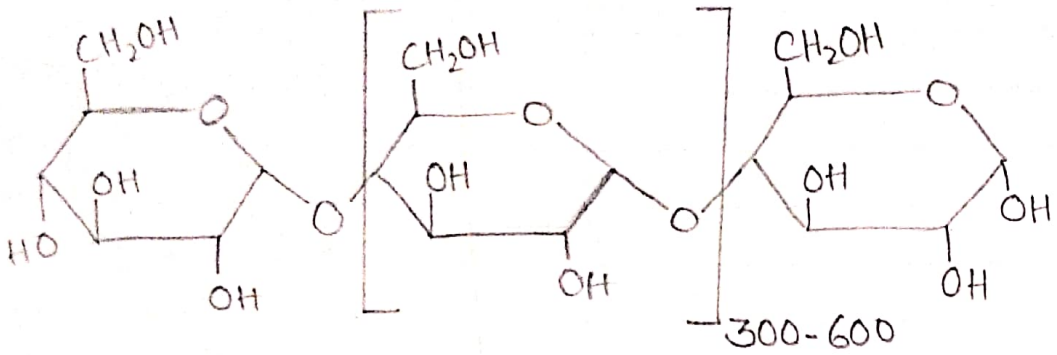


Fig- Amylose – $(C_6H_{10}O_5)_n$

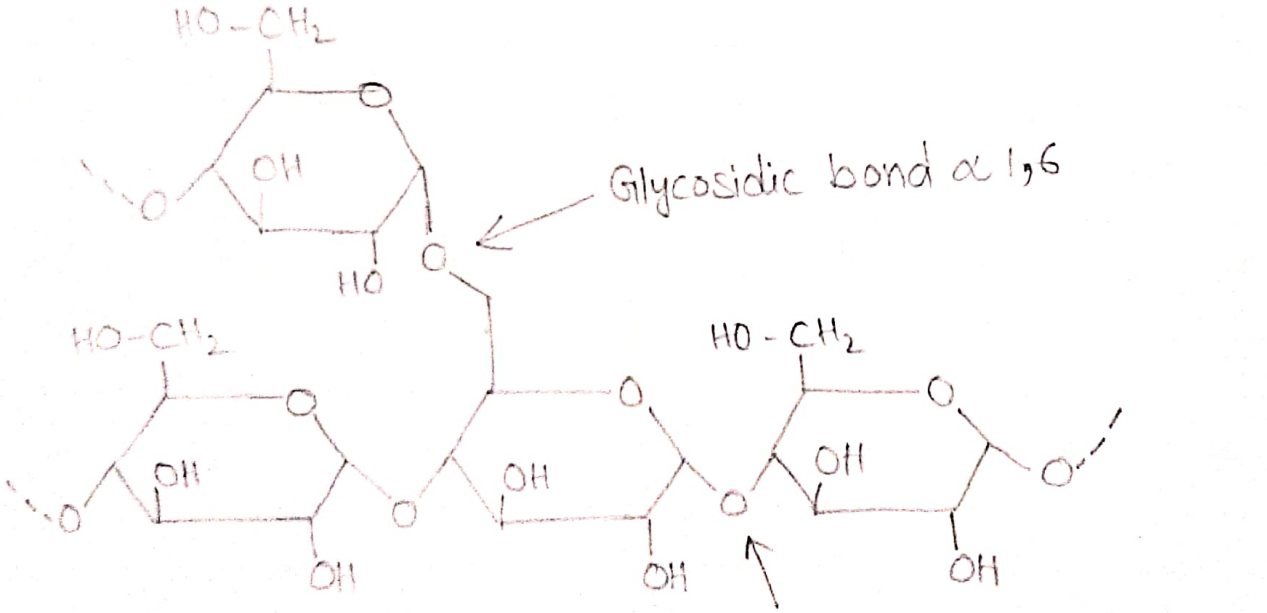


Fig- Amylopectin

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Expt. No. _____ Page No. 1

MICROSCOPIC STUDY OF STARCH GRAINS

Introduction:

Starch grains are the common insoluble complex of carbo-hydrate and are the most important reserve food for future use of plant. They are mainly found in storage organ of potato tubers, gram seed, banana seed, maize grain & pea seed. Starch grains may be found in all parenkymatic tissue mainly in tuber, rhizome, from in all parenkymatic tissue in colourless plastid namely ~~amyloplast~~ aminoplast. Chemically starch is made up of long chain of anhydrous glucose resibues. They are from in colourless plastid. They are mainly composed of two component amylase and amylopectine. They are different size and shape. Some large enough to be seen by naked eye but most are microscopic.

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Every starch grain has a shiny refractive point called hilum (the starting point). Starch molecules are diposided in layers. Around the hilum giving rise to a disting stratified appearance. This layers is called lives of stratification. According to the position of hilum starch grains are of two types-

i) Ecentric: When hilum are situated at one aim on the grain. Eg- Potato strain.

ii) Concentric: When the hilum concentrally situated in the grain. Eg- Pea, Wheat.

Starch grains may be simple, compound or semicompound.

i) Simple starch grain:-
Starch grains are free and seperate from each other. Eg- Potato.

ii) Compound starch grain:-
When two or more grains are united with each other, thus two or more hila. Eg- rice.

iii) Semicompound starch grain:-
Compound starch grain sorrounded by common line of stratification. Eg- Metroxylon sago.

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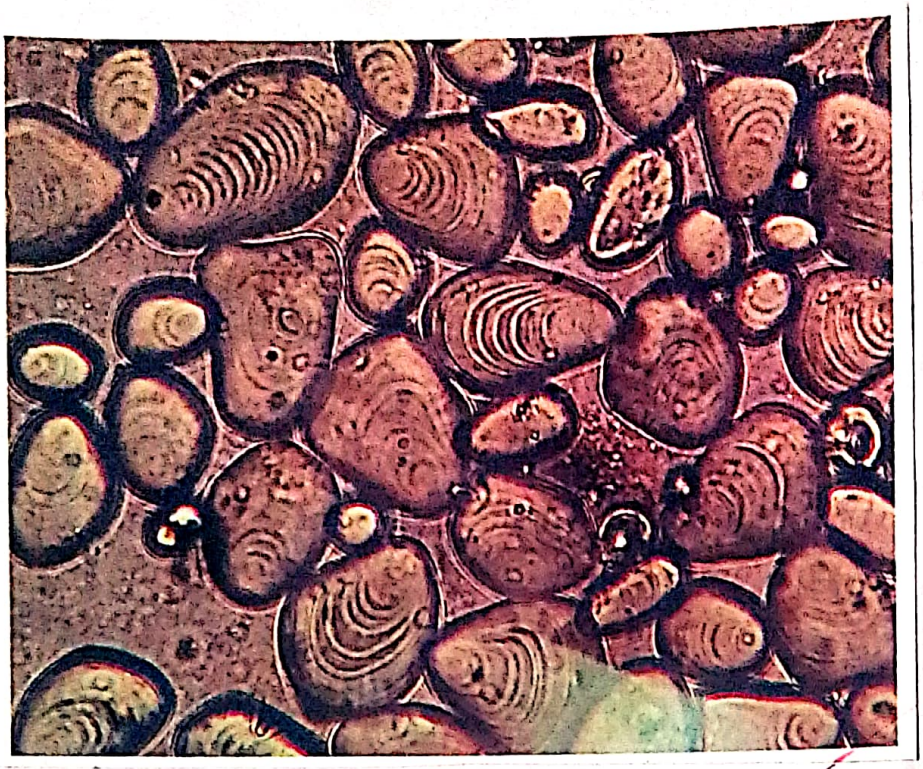


Fig- Micrascopic view of starch grains of potato tubers (print)

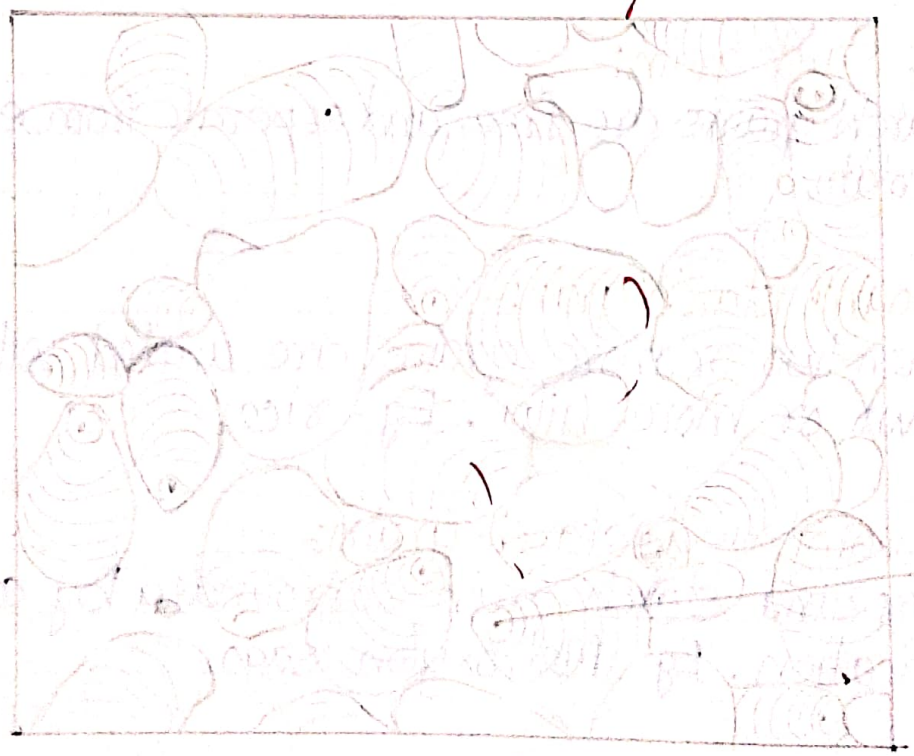


Fig- Study of starch grains of potato tubers (sketch)

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Expt. No. _____ Page No. 3

Study of starch grains of potato tubers

Requirements:

i) Potato tuber ii) Knife iii) Scalpel iv) Compound microscope v) Slide vi) Cover slip vii) Water viii) 1% iodine solution ix) Bloating paper.

Method:

i) Cut the potato in half and scrape a sittle of the potato on the glass slide, there should not be any potato peaches on the slide.

ii) Put a small drop of 1% iodine solution on the potato juice and then a cover slip are place on the top.

iii) Use bloating paper to wipe out any extra iodine solution from the top.

iv) The slides are observed under the compound microscope.

Observation:

i) The potato starch grains are tend to be oval in shape.

ii) They react with the iodine solution and turned into blue and react black in colour.

iii) They are larger, typically 20-60 um in longest dimention.

iv) The grains are simple to semicompound in nature with ecentric hilum.

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